**Paper return** is the difference between the ending portfolio value and its starting value evaluated at the manager's decision price. This is:

$$Paper\ Return = S \cdot P_n - S \cdot P_d \tag{3.2}$$

Here $S$ represents the total number of shares to trade, $P_d$ is the manager's decision price, and $P_n$ is the price at the end of period n. $S \cdot P_d$ represents the starting portfolio value and $S \cdot P_n$ represents the ending portfolio value. Notice that the formulation of the paper return does not include any transaction costs such as commissions, ticket charges, etc. The paper return is meant to capture the full potential of the manager's stock picking ability. For example, suppose a manager decides to purchase 5000 shares of a stock trading at \$10 and by the end of the day the stock is trading at \$11. The value of the portfolio at the time of the investment decision was \$50,000 and the value of the portfolio at the end of the day was \$55,000. Therefore, the paper return of this investment idea is \$5000.

**Actual portfolio return** is the difference between the actual ending portfolio value and the value that was required to acquire the portfolio minus all fees corresponding to the transaction. Mathematically, this is:

$$Actual\ Portfolio\ Return = \left(\sum s_j\right) \cdot P_n - \sum s_j p_j - fees \tag{3.3}$$

where,

$(\sum s_j)$ represents the total number of shares in the portfolio

$(\sum s_j) \cdot P_n$ is the ending portfolio value

$\sum s_j p_j$ is the price paid to acquire the portfolio

and fees represent the fixed fees required to facilitate the trade such as commission, taxes, clearing and settlement charges, ticket charges, rebates, etc. $s_j$ and $p_j$ represent the shares and price corresponding to the $j^{th}$ transaction.

For example, suppose a manager decides to purchase 5000 shares of stock trading at \$10. However, due to market impact, price appreciation, etc., the average transaction price of the order was \$10.50 indicating that the manager invested \$52,500 into the portfolio. If the stock price at the end of the day is \$11 the portfolio value is then worth \$55,000. If the total fees were \$100, then the actual portfolio return is \$55,000 − \$52,500 − \$100 = \$2400.